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Macro Economics

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1.2.1. The Keynesian Macroeconomics

- Macroeconomics was not formally born until the **Great Depression (1929 – 1933)**, a decade of high unemployment and stagnant production.
- The market adjustment concept of the classical and neoclassical schools didn't work at the period, and it was as a result of this failure that Keynes developed expansionary policies.
- The expansionary policies consist of two elements:



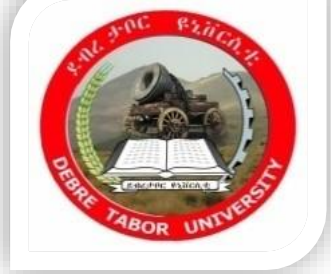
The Keynesian Macroeconomics

- A) *Increasing money supply* - Increase in money supply decreases the interest rate.
- This encourages agents (investors) to invest their money rather than putting it in banks. The increase in investment results in increased output
 - However, interest rate (and thus investors) may not respond to the change in money supply or all the increase in money supply may be absorbed at the existing interest rate.
 - This is particularly the case when recession is deep. If this is the case, the economy is said to be in a *liquidity trap*.
 - If there is liquidity trap, Keynes argued that, the Classical model will be incapable of producing an equilibrium, and that monetary policy is ineffective.



The Keynesian Macroeconomics

- In such a case, the second element is proposed.
- *ii. Increasing government expenditure* - so as to increase aggregate demand, which will in turn encourage production. Increased output/income again increases consumption, consumption further boosting aggregate demand ... – ***the multiplier effect***.
- Keynes focused primarily on the short-term. He wanted to cure an immediate problem almost regardless of the long-term consequences of the cure.



The Keynesian Macroeconomics

- Keynes focused primarily on the short-term.
- However, increased government spending:
 - might trigger inflation
 - might lower the long-term growth rate of production (the economy would create fewer jobs and thus unemployment rate would rise).
- For the Keynesians, inflation can be kept under control with either a **contractionary fiscal or a contractionary monetary policy**.
- Their basic assumptions are:
 1. Inflexible nominal wages and prices in the downward direction
 2. The economy is unstable subject to shifts in AD (Aggregate Demand),
 3. There is a large multiplier effect for changes in government spending and tax rates.

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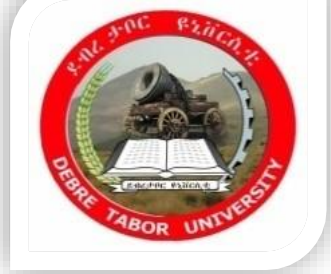
Chapter Two

THE NATIONAL INCOME ACCOUNTING



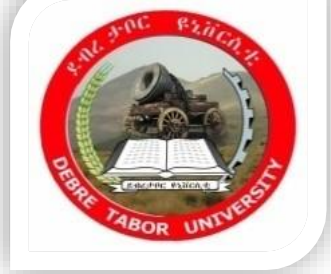
Introduction

- National income accounting
 - ❖ designed to measure the overall production performance of the economy
 - comparing the national income accounts over a period of time
 - provides a basis for the formulation and application of public policies
- It is generally agreed that the best available indicator of an economy's health (wellbeing) is **its total annual output of goods and services, or the economy's aggregate output**
- The basic social accounting measures of the total output of goods and services are Gross Domestic Product (GDP) and Gross National Product (GNP).



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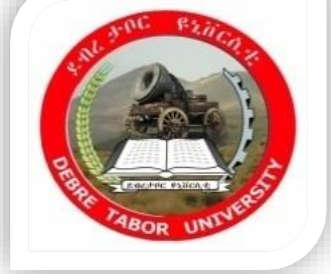
- **GDP** is defined as the total monetary value of *all final goods and services* produced in the *territories (within the boundary)* of the economy in a given year.
- **GNP** is defined as the total monetary value of *all final goods & services* produced by *citizens* of the country in a given year.
 - $GNP = GDP + \text{Net Factor Income (NFI)}$
 - **NFI** = Factor income generated from citizens (all resources) living abroad minus factor income flowing out by foreigners living in host country.
- **GDP** is more commonly taken as the basic measure of a nation's output as compared to GNP.
 - GDP is easier to measure
 - GDP is the better measure of the job-creating potential of an economy than is GNP
 - It makes international comparisons easier, as most countries use GDP.



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- GDP ignores transactions involving intermediate goods (in order to avoid double counting).
 - National income accountants are careful to calculate only the value added (**the market value of a firm's output less the value of the inputs, which it has purchased from others**) by each firm.
- GDP also excludes two non-productive transactions i.e.
 - ***Purely financial transactions*** (Public transfer payments, Private transfer payments, Buying and selling of securities). Because recipients make no contribution to current production in return for them.
 - ***Second hand sales***: because such sales either reflect no current production, or they involve double counting.

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2.1. Approaches to Measuring GDP

1. Product (Output) Approach or Value Added Approach
2. Expenditure Approach
3. Income Approach

1. Product (Output) Approach or Value Added Approach

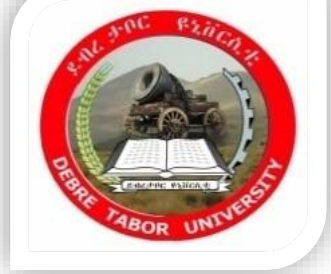
- GDP can be obtained either by:
 - taking the market value of final goods and services or
 - taking the value added at each stage of production.

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Stages of Production	Sales value of product	Value Added
Firm A, sheep ranch	\$ 60	\$60
Firm B, Wool processor	100	40
Firm C, suit manufacturer	125	25
Firm D, clothing wholesaler	175	50
Firm E, clothing retailer	\$250	75
Total sales value	\$ 710	
Value added (total income)		\$250

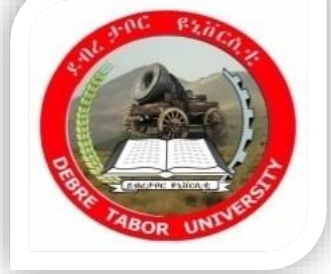


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- by calculating and summing the values added by all firms (sectors) in the economy, we can determine the GDP, that is, monetary value of the total output.

2. Expenditure Approach

- GDP can also be determined by adding up all that is spent on this year's total output (Expenditure-side)
- All final goods produced in an economy are purchased either by the three domestic sectors:
 - Households(C).....C represents Personal consumption expenditure
 - Government(G)
 - Business enterprises(I)..... I represents *Gross domestic private investment*
 - by foreign nations(X-M)



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- *Gross domestic private investment (I)*: replacement and added investment.
- replacement investment which implies depreciation (capital used up), D , and added investment that is known as net investment (**In**). Thus, $I = D + In$.

$$GDP = C + I + G + X_n$$

3. Income Approach:

- by summing up all the incomes derived from the production of this year's output.
- By the money income derived from the production of this year's output (Income-side)
- This approach measures GDP in terms of income earned



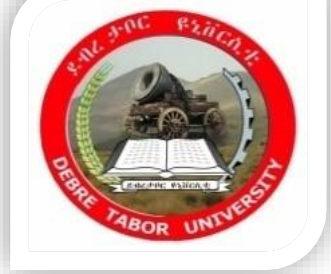
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- It is the sum of all incomes received from all factors of production which contribute to the production process plus two additional non-factor payments.

- The main income categories are:

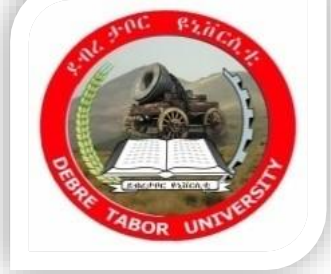
$$GDP = (W+S) + r + i + \Pi + D + IBT$$

- $(W+S)$ is **Compensation of employees**. It comprises wages and salaries paid by governments and businesses.
- r (**Rents**): Consists of income payments received by households and businesses, which supply property resources.



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- **Interest (i):** comprises items such as the *net* interest payments households receive on saving deposits, certificate of deposits (CDs) and corporate bonds.
- **Profit(Π)= Proprietor's income or profit (Π_P)+ Corporate Profits (Π_C)**
- **Proprietor's income or profit (Π_P):** is net income of sole proprietorships and partnerships (or income of unincorporated businesses).
- **Corporate Profits (Π_C) :** may be divided into three:



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- may be collected as corporate income taxes.
 - may be distributed as dividends (to stockholders).
 - may be retained as undistributed corporate profits).
- $IIC = \text{corporate income tax} + \text{dividend} + \text{undistributed corporate profits}$).
 - Adding employees' compensation, rents, interests, Π_p & Π_c , and NFI we get a country's National Income (**NI**): $NI = a + b + c + d + e + NFI$.
 - **Indirect Business Taxes (IBT)**: Which firms treat as costs of production, and therefore, add to the prices of the products they sell. E.g. sales tax, excise tax, business property tax, license fee.

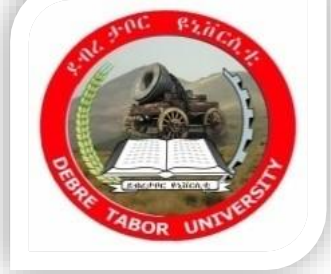
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- **Consumption of fixed capital (depreciation, D):** The annual charge, which estimates the amount of capital equipment used up in each year's production, is called **Depreciation**.

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2.2. Other Social Accounts (GNP, NNP, NI, PI and DI)

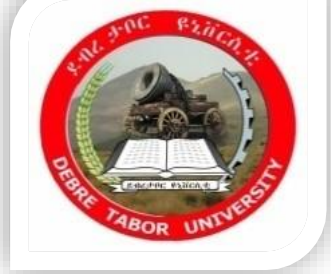
1. **GNP** = GDP + NFI

2. **Net National Product (NNP)**: Is GNP adjusted for depreciation changes.

$$NNP = GNP - D$$

3. **National Income (NI)**: The only component of NNP that does not reflect the current productive contributions of economic resources is IBT. Thus

$$NI = NNP - IBT$$



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4. **Personal Income (PI):** is income received by households.

$PI = NI$ (income earned)

- Social security contribution
- Corporate income taxes
- Undisturbed corporate profits
- + Transfer payments

❖ These are not actually received by households and should be deducted if received but not currently earned.

5. **Disposable Income (DI):** the income individuals have to spend or save after payment of taxes.

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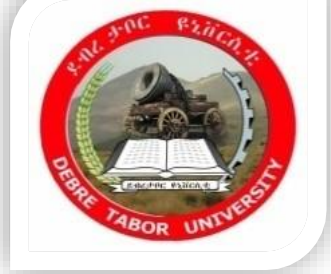
- It is simply personal income less personal tax and non-tax payments (PT).

$$\mathbf{DI = PI - PT}$$



2.3. Nominal versus Real GDP

- the market value of all final goods and services at current prices.
- GDP equals the sum of $P_i Q_i$
- **Real (constant-Birr) GDP** measures each year's output in terms of the prices prevailed in a selected **base year**



2.4. The GDP Deflator and the Consumer Price Index

1. GDP deflator= $\text{Nominal GDP} / \text{Real GDP} * 100$

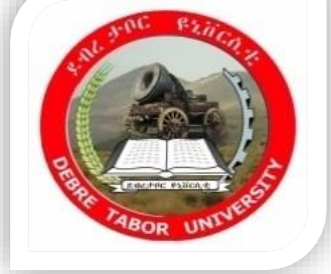
- measures the price of output relative to its price in base year.
- GDP deflator reflects what is happening to the overall level of prices in the economy.

2. *The Consumer Price Index (CPI):*

- CPI is the most commonly used measure of the level of prices (or of the cost of living).

$$\text{CPI} = \frac{\text{Cost of a Market Basket of Products at Current Prices}}{\text{Cost of the Same Basket of Products at Base Year Prices}} * 100$$

Note that $\text{CPI} = 100$ for the selected base year.



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- CPI differs from the GDP deflator in three main ways:
 1. GDP deflator measures the prices of a much wider (all) group of goods and services than CPI does.
 - The CPI measures only the prices of goods and services bought by consumers.
 - An increase in the prices of goods and services bought by firms or the government will show up in the GDP deflator but not in the CPI.
 2. The CPI measures the cost of a given basket of goods and services, which is the same from year to year.
 - The basket of goods and services included in the GDP deflator, however, differs from year to year, depending on what is produced in the economy in each year.

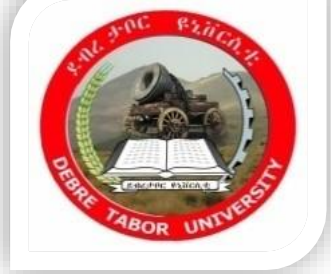


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3. The CPI directly includes prices of imports, whereas the GDP deflator includes only prices of products produced domestically.

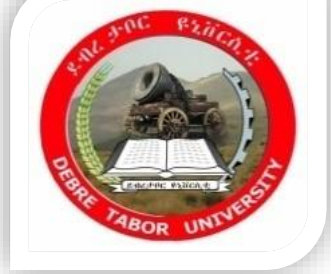
- Neither of these two price indices is clearly superior to the other in measuring the cost of living.
- Thus, the CPI is also used to deflate nominal GDP so as to arrive at the real GDP.

$$\text{Real GDP} = \text{Nominal GDP} / \text{CPI} * 100$$



2.5. GDP and Welfare

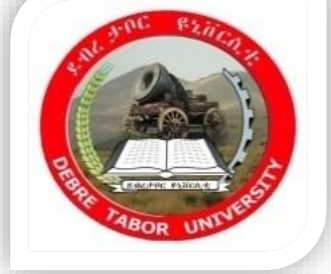
- GDP has some limitations in gauging the social wellbeing of the people in a nation.
 1. Non-market transactions are not included.
 2. It doesn't include the underground economy (*black market*) transactions
 3. It ignores the quality aspect of goods and services.
 4. It does not consider the cost of environmental damage
 5. The satisfaction that one gets from recreational activities and other uses of his/her leisure time are not accounted for.
 6. It is based on estimations and thus may not accurately reflect even the transactions it explicitly intended to measure.



2.6. The Business Cycle

$$\text{Output Gap} = \text{Potential Output} - \text{Actual Output}$$

- Output gap grows during a recession
- The gap declines and ultimately even becomes negative during an expansion.
- A negative gap means that there is over employment, overtime for workers, and more than the usual rate of utilization of machinery.



2.7. Unemployment and Inflation

2.7.1. Unemployment

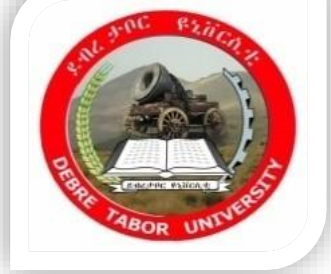
- ❖ Total population
 - working-age Vs outside the working-age
- ❖ Working-age
 - currently active and currently inactive
- ❖ People in the working-age category who are either employed or actively seeking a job constitute the labor force.
 - Labour force= employed + unemployed



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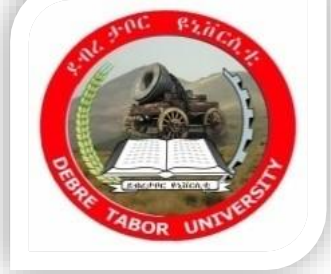
- A person is said to be unemployed if he/she is in the working-age, available for work, actively seeking work, but doesn't have one
 - ❖ The unemployment rate=
$$\frac{\text{Number of unemployed}}{\text{Labour force}} \times 100$$
- **Types of Unemployment:**
 - Frictional Unemployment
 - Structural Unemployment
 - Cyclical Unemployment

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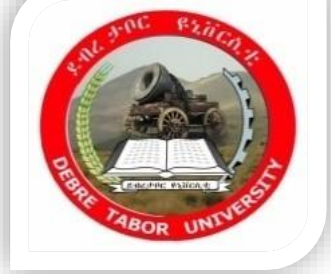
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- Full employment does not mean zero unemployment.
- It refers to the situation that occurs when the actual rate of unemployment is no more than the natural rate of unemployment.
- In other words, the natural rate of unemployment is the sum of the frictional and structural un employments expected over some period (usually a year).
- An economy in which the actual unemployment rate is less than the natural rate of unemployment is termed as an **overheated economy**
- the economy can produce more than the potential real GDP, implying that the economy's capacity output exceeds the potential real GDP.



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- Cyclical Unemployment (which is usually characterized by *layoffs* – temporary suspensions of employment without pay) tends to increase during contractions (recessions).
- This negative relationship between changes in real GDP and changes in the unemployment rate is known as Okun's law.
- The main costs of unemployment are:
 - Output is lost (GDP falls)
 - Distortional impact
 - The unemployed may have more leisure when not working. But this benefit is more than offset by the costs to the society (unwanted leisure has little satisfaction, the gov't loses tax revenue → negative effect on the society).



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2.7.2. Inflation:

- When inflation exists, the purchasing power of a nation's currency declines over time.

$$\text{Inflation Rate at Period } t = (\text{CPI}_t - \text{CPI}_{t-1}) / \text{CPI}_{t-1} * 100$$

- **Pure Inflation:** refers to a condition that occurs when the prices of all goods rise by the same percentage over the year.
 - no changes in the relative prices of goods.
 - does not provide consumers with any incentive to substitute one good with the other in their budget
- **Hyper Inflation:** inflation at very high rates prevailing for at least one year.



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Effects of Inflation:

1. redistribution of income and wealth from creditors to debtors. Debtors can pay back loans in currency units that have less purchasing power than what they borrowed. It can also harm savers.
2. Hyperinflation seriously impairs the functioning of the economy by causing credit markets to collapse and by wiping out the purchasing power of accumulated savings.
3. Actions taken in anticipation of inflation can adversely affect the performance of the economy.
4. Anticipated inflation can distort consumer choices by causing buyers to purchase goods now that they might otherwise prefer to purchase in the future.



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- Expansionary aggregate demand policies tend to produce inflation, unless they occur when the economy is at high levels of unemployment.
- Inflation could be **demand-pull** inflation (where high aggregate demand is responsible for it) or **cost-push** inflation (where adverse supply shocks are typically events that push up the costs of production).



2.7.3. The Phillips Curve

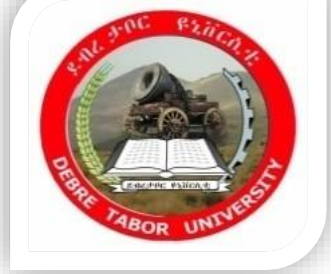
- Building the Phillips Curve:

$$\pi = \pi^e - \beta(u - u^n) + v$$

The Phillips curve states that inflation depends on expected inflation...

the deviation of unemployment from the natural rate (cyclical unemployment)...

and supply shocks.



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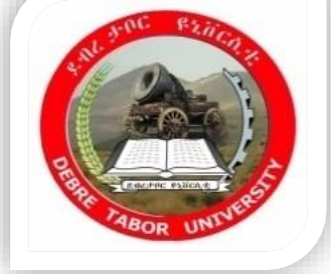
- A simple and plausible assumption might be that people form expectations about future inflation based on recent inflation.

$$\pi^e = \pi_{-1}$$

$$\pi = \pi_{-1} - \beta(u - u^n) + v$$

- In this case, we can write the Phillips curve as...
- states that inflation depends on past inflation, cyclical unemployment, and a supply shock.
- The first term in the Phillips curve implies that inflation has inertia and that inflation keeps going unless something acts to stop it. In essence we have inflation because we expect it and we expect it because we have it.

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- The Phillips curve describes an empirical relationship between inflation and unemployment: the higher the rate of unemployment, the lower the rate of inflation.
- The curve suggests that less unemployment can always be attained by incurring more inflation and vice versa.
- In other words, the curve suggests a trade-off between inflation and unemployment.



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- **Two Causes of Rising and Falling Inflation**
- The second term shows that cyclical unemployment exerts upward or downward pressure on inflation. Low unemployment pulls inflation up.
- This is called **demand-pull inflation** because high AD is the cause.
- The third term shows that inflation also rises and falls with supply shocks.

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- An adverse supply shock would push production prices up. This type of inflation is called **cost-push inflation**.
- This trade-off between unemployment and inflation holds in the short run.
- In the long run, the unemployment rate is basically independent of the long run inflation rate.